

Assignment 5 - RSA Encryption & Decryption

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No.: 29

$$\begin{aligned} M &= \\ &= 81 + \text{mod}(404129 * 19, 400) \\ &= \end{aligned}$$

Encryption: $C = M^e \text{ mod } n$

Decryption: $M = C^d \text{ mod } n$

Where

$$\begin{aligned} n &= 481, \\ e &= 17 = 0b10001, \\ d &= 305 = 100110001 \end{aligned}$$

In my case $M = 132$

Therefore

$$\begin{aligned} C &= M^e \text{ mod } n = 132^{17} \text{ mod } 481 \\ C &= 474 \end{aligned}$$

Trace:

Step	Exponent	Binary	Bit	Base
1	17	10001	1	132
2	8	1000	0	108
3	4	100	0	120
4	2	10	0	451
5	1	1	1	419

And decrypting back to plain text
 $M = C^d \bmod n = 474^{305} \bmod 481$
 $M = 132$

Trace:

Step	Exponent	Binary	Bit	Base
1	305	100110001	1	474
2	152	10011000	0	49
3	76	1001100	0	477
4	38	100110	0	16
5	19	10011	1	256
6	9	1001	1	120
7	4	100	0	451
8	2	10	0	419
9	1	1	1	477